

Equilibrium Distribution for t-Distributed Stochastic Neighbor Embedding with Generalized Kernel

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1 Introduction

The t-distributed stochastic neighbor embedding(t-SNE) is a well-known algorithm for dimension reduction and data visualization. Let $d, s \geq 2$ with $d > s$, and $n \geq 1$. The input of t-SNE consist of vectors $X_1, \dots, X_n \in \mathbb{R}^d$. t-SNE aims to find vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$ that serve as a good representation of the original vectors. The process starts with defining the

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following probability masses:

$$p_{j|i} = \frac{\exp(-\|X_i - X_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|X_i - X_k\|^2 / 2\sigma_i^2)}, \quad (1.1)$$

where σ_i are given constants. For each pair of data points X_i and X_j , we assign a weight to that pair, namely the Gaussian function $\exp(-\|X_i - X_j\|^2 / 2\sigma_i^2)$. We call such function the *input kernel*. In this paper, we are looking at generalized kernels. Let $\mathbf{Ker}_I(x, y, \sigma) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ be such that $\mathbf{Ker}_I(x, y, \sigma) = \mathbf{Ker}_I(y, x, \sigma)$. We replace the probability mass in (1.1) with the following

$$p_{j|i} = \frac{\mathbf{Ker}_I(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker}_I(X_i, X_k, \sigma_i)}. \quad (1.2)$$

σ_i satisfies that for each $i = 1, \dots, n$, that

$$-\sum_{j=1, j \neq i}^n p_{j|i} \log p_{j|i} = \log \mathbf{Perp} =: \log(n\rho). \quad (1.3)$$

$\mathbf{Perp}$ is called the perplexity of the model and $\rho \in (0, 1)$ is a given constant. The purpose of σ_i is to make sure that for each i , the probability distribution induced by $p_{j|i}$ has the same entropy across all i . Clearly, $p_{j|i}$ is not symmetric. We then define $p_{ii} = 0$ and for $1 \leq i \neq j \leq n$,

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} = \frac{1}{2n} \left(\frac{\mathbf{Ker}_I(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker}_I(X_i, X_k, \sigma_i)} + \frac{\mathbf{Ker}_I(X_i, X_j, \sigma_j)}{\sum_{k \neq j} \mathbf{Ker}_I(X_j, X_k, \sigma_j)} \right). \quad (1.4)$$

Obviously,

$$\sum_{\substack{j=1 \\ j \neq i}}^n p_{j|i} = 1, \quad \sum_{1 \leq i \neq j \leq n} p_{ij} = 1. \quad (1.5)$$

As for the output vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$, set for $1 \leq i \neq j \leq n$,

$$q_{ij} = \frac{\mathbf{Ker}_O(Y_i, Y_j)}{\sum_{k \neq l} \mathbf{Ker}_O(Y_k, Y_l)}. \quad (1.6)$$

The weight we assign to each pair of output data points Y_i and Y_j , a.k.a. a function $\mathbf{Ker}_O(Y_i, Y_j) : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}^+$, is called the output kernel. In the original algorithm, $\mathbf{Ker}_O(Y_i, Y_j) = (1 + \|Y_i - Y_j\|^2)^{-1}$. Denote $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ the input and output data set. The goal of t-SNE is to find Y that minimizes the relative entropy

$$L_{n,\rho}(X, Y) = \sum_{1 \leq i \neq j \leq n} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right). \quad (1.7)$$

More precisely, the output of t-SNE Y^* is defined as

$$Y^* = \arg \min_{Y \in (\mathbb{R}^s)^n} L_n(X, Y). \quad (1.8)$$

In the expression of $L_n(X, Y)$, all p_{ij} are given and q_{ij} is a function of Y_i and Y_j . Therefore, Y^* can be found simply by using gradient descent. This is no different from the original version of t-SNE. However, it is the mathematical properties and implications of t-SNE that we are interested in. Towards this end, let $\mathcal{M}(\mathbb{R}^d)$ be the space of Borel probability measures on $\mathbb{R}^d$ and $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ be a probability measure. Suppose all input

$X = (X_1, \dots, X_n)$ are i.i.d., independent random variables with probability measure μ_X . Meanwhile, let $Y^* = (Y_1, \dots, Y_n)$ be the output of the t-SNE algorithm. Consider the following empirical measure:

$$\mu_n := \frac{1}{n} \sum_{i=1}^n \delta(X_i, Y_i) \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s). \quad (1.9)$$

We are interested in its limiting measure as $n \rightarrow \infty$. To finish the setup, there are a few more remaining quantity we would like to introduce. Let $\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s)$. For a given kernel $\mathbf{Ker}_I(x, y)$, define $F_{\rho, \mu}(x, \sigma) : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}$ as

$$F_{\rho, \mu}(x, \sigma) = \int \frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho. \quad (1.10)$$

Note that the last s variables in the integrands stay constant. It is not hard to see that $F_{\rho, \mu}(x, \sigma)$ is the continuous version of the entropy. Next, we extend the definition of σ_i by defining $\sigma_{\rho, \mu}^* : \mathbb{R}^d \rightarrow \mathbb{R}$ the unique solution of the equation

$$F_{\rho, \mu}(x, \sigma) = 0. \quad (1.11)$$

The introduction of σ^* aligns with the procedure of finding σ_i that controls the entropy to be the same for all i . Of course, the existence and uniqueness of σ^* is not trivial and will be discussed in section 6. For the symmetric probability mass, let $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a given function and $p_\psi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be defined by

$$p_\psi(x, x') = \frac{1}{2} \left(\frac{\mathbf{Ker}_I(x, x', \psi(x))}{\int \mathbf{Ker}_I(x, x', \psi(x)) d\mu(x')} + \frac{\mathbf{Ker}_I(x, x', \psi(x'))}{\int \mathbf{Ker}_I(x, x', \psi(x')) d\mu(x)} \right). \quad (1.12)$$

For the extension on the output side, let $q : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}$ be given by

$$q(y, y') = \frac{\mathbf{Ker}_O(y, y')}{\iint \mathbf{Ker}_O(y, y') d\mu(y) d\mu(y')}. \quad (1.13)$$

Based on the construction of p and q , define

$$I_\rho(\mu) = \iint p_{\sigma_{\rho, \mu}^*}(x, x') \log \left(\frac{p_{\sigma_{\rho, \mu}^*}(x, x')}{q(y, y')} \right) d\mu(x, y) d\mu(x', y'). \quad (1.14)$$

Recall $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ is an arbitrary given probability measure. We set $\mathcal{P}_X$ to be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^s$ whose marginal coincides with μ_X . Formally speaking,

$$\mathcal{P}_X := \{\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s) : \text{the } \mathbb{R}^d \text{ marginal of } \mu \text{ is } \mu_X\}. \quad (1.15)$$

Lastly, define

$$\tilde{\mathcal{P}}_X := \left\{ \mu \in \mathcal{P}_X : \int (p_{\sigma_{\rho, \mu}^*}(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} \mathbf{Ker}_O(y, y')}{\mathbf{Ker}_O(y, y')} d\mu(x', y') = 0, \mu(x, y)\text{-a.e.} \right\}. \quad (1.16)$$

2 Main Results

The core results of this problem are the conditions we assume the input and output kernels to have such that the output of the generalized t-SNE algorithm converges to a limiting equilibrium measure. Towards this end, below are the constraints we impose on the input and output kernels.

Condition 1 (Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of*

$$\mathbf{Ker}_I(x, x', \sigma) = \exp(-w(\sigma \|x - x'\|^\theta)) \quad (2.1)$$

Here, σ is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. Then w satisfies:

1. $\forall t \geq 0, w'(t) > 0. \lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. *The following integral converges:*

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (2.2)$$

Since the all probability masses are normalized, we can assume without loss of generality that $w(0) = 0$ from now on.

Condition 2 (Output Kernel Conditions). *The output kernel depends only on the distance between two output data points y and y' . In particular, $\mathbf{Ker}_O(y, y') = k(\|y - y'\|)$ and that:*

1. $k : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^+$ is decreasing and bounded;
2. $\mathbf{Ker}_O$ satisfies

$$\iint_{(0, \infty)^2} \mathbf{Ker}_O(y, y') dy dy' < \infty; \quad (2.3)$$

3. k has bounded derivative;
4. $k'(0) = 0$.

We are ready to state the main results. Note that compared to [Auffinger and Fletcher \[2023\]](#), we no longer require μ_X to have C^1 density. We managed to relax the condition to C^0 .

Theorem 1. *Assume μ_X has a continuous density $f(x)$ and compact support. Let $X_i \in \mathbb{R}^d$ be i.i.d random variables with probability measure μ_X . If condition 1 and condition 2 hold, then for any $\rho \in (0, 1)$,*

$$\lim_{n \rightarrow \infty} \inf_Y L_{n, \rho}(X, Y) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu). \quad (2.4)$$

Moreover, there exist a measure μ^ and a sub-sequence $\{n_k\}$ such that $\mu_{n_k} \rightarrow \mu^*$ weakly and $I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$.*

Theorem 2. *Under the same assumptions of theorem 1, μ^* has compact support.*

From section 7 up to a very late point in our proofs, we will prove our lemmas under lighter assumptions on the probability measure μ_X we draw the input samples from. We will assume μ_X is sub-Gaussian rather than with compact support. For results in those parts to be true, we need to add one more constraint to the input kernel, as shown in the **condition below**. The reader can easily verify that if we assume μ_X to be compact from the beginning, all the lemmas can be easily proven using the exact same framework without the extra condition below (and in fact, easier). We will discuss why we choose to prove those lemmas under the sub-Gaussian assumption, as well as other intuition and insights in section 3. We will also reiterate how the proofs are structured in section 4.

Condition 3 (Alternative Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_I(x, x, \sigma) = \exp(-w(\sigma \|x - x'\|^\theta))$. Here, σ is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. Define*

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0, \text{ s.t. } \lim_{n \rightarrow \infty} n^{-1/2} \exp \left(6w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = O(n^{-\epsilon}) \right\}. \quad (2.5)$$

Then w satisfies:

1. $\forall t \geq 0, w'(t) > 0. \lim_{t \rightarrow \infty} w(t) = \infty;$
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0;$
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (2.6)$$

4. $\exists M > 0$ s.t. $\forall t \geq 0, w'(t) \leq w(t)$, and $\sup \mathcal{K}_w > 0$.

In short, if a lemma can be proved with assuming condition 3 and μ_X being sub-Gaussian, then it can also be proved in a very similar (and most likely easier) way by assuming only condition 1 and μ_X being compactly supported.

3 Intuitive Insights

We use this section to provide some intuition and insights on the context. From the form of the kernels we choose, to why we introduce the extra condition 3, and to a series of simulation plots that demonstrate the effectiveness of the generalized t -SNE algorithm. We hope to provide the reader with a better understanding of the problem and the results we aim to prove.

Is the form of the input kernel general enough?

The input kernel is the most important part of the algorithm. Since we would like to expand the result of the original version, it is important that we do not lose comprehensiveness when assuming a concrete form of the input kernel. Therefore, (2.1) is due an explanation.

We start by realizing that t -SNE is about the relationship between all data points. The insights we gain from the algorithm should be translation and rotation invariant with respect to the coordinate system we choose. Hence, it is only reasonable to assume that the input kernel is a function of $\|x - x'\|$ for each pair of input data points x, x' .

On the other hand, the weight we assign to each pair of data points is always positive due to nature of probability mass. We thus lose no generality by assuming the input kernel has an exponential form because obviously, the exponential function is a smooth bijection from $\mathbb{R}$ to $\mathbb{R}^+$.

Additionally, though it is coincidental that in the original version of the algorithm σ_i can be viewed as the variance of a Gaussian random variable, in reality σ_i 's are only used to control the entropy of each probability distribution induced by x_i . With that being said, we only need to introduce σ in such a way that it is able to control the entropy for each x . As we can see in section 6, simply multiplying σ to $\|x - x'\|^\theta$ is enough to achieve this goal. Moreover, we prove the convergence of σ^* in section 7, providing the anchor of the following results just as in [Auffinger and Fletcher \[2023\]](#).

Finally, combining all the above, we arrive at the form of the input kernel in (2.1), by adding a general function w to the product $\sigma \|x - x'\|^\theta$. This gives us a kernel we can easily work without losing any generality.

Are the conditions on the input/output kernel general enough?

Besides the form of the input kernel, we also want to make sure that the conditions we impose on the kernels are also general enough for the significance of our results. Since we frequently deal with integrals containing the input and output kernels, their integrability is the bare minimum we should assume. Along with integrability comes their decay to 0 at infinity. Apart from that, we do not add any excessive constraints to the kernels and readers will see later in proofs that those assumptions are strictly necessary for the results to hold.

Specifically, notice the second assumption of condition 1 is basically requiring the input kernel to decay faster than a minimal speed by controlling its second order derivative. If we let $w(t) = \log t$, we have $w'(t) + tw''(t) = 0$. Roughly speaking, this implies that the main theorems hold as long as the input kernel decays at polynomial rate or faster, which is a big step up than proving the convergence for Gaussian kernels alone.

Why do we introduce condition 3?

[Yi: Ask Tuca about this and come back. Tuca said the convergence for σ^* is "important in its own right". I would like to write a little bit more about this.]

4 Structure of the Proofs

The outline of the proofs should be self-explanatory by the titles of each sections. However, we would still like to dedicate this short section to remind the readers of how the proofs are structured. We have two conditions for the input kernels, i.e., Condition 1 and Condition 3 and latter has one more assumption compared to the former while being the same for everything else.

Since the convergence of components are important in their own rights, as we discussed in section 3, we will prove lemmas assuming μ_X is sub-Gaussian and condition 3 when we are able to. All proofs can be easily adapted to the case where μ_X is compactly supported and condition 1 is assumed. To summarize, the flows of the proofs will not be disrupted by proving lemmas under the sub-Gaussian assumption with condition 3. We will also indicate which assumptions we are using in each proof to avoid confusion.

5 Simulations

[Yi: Under Construction] Another important question we would like to ask, is that can the results in the main theorems be visualized and thus verified? In theory, if we choose any input and output kernels meeting condition 1 and 2 respectively, we are able to get an asymptotic equilibrium measure and each equilibrium measure will look different for different combinations of input and output kernels.

Towards that end, we choose four different combinations of input and output kernels, including the original one, and run t -SNE algorithms on the exactly same subset of the famous MNIST data set with exactly same settings, apart from different kernels of our choices. As

shown in figure 1, we used four different configurations for the input and output kernels. They are

- (a) Original t -SNE kernels;
- (b) Polynomial input kernel $\mathbf{Ker_I}(x_i, x_j, \sigma_i) = (\sigma_i \|x_i - x_j\|^{10} + 1)^{-1}$;
- (c) Mixed input kernel $\mathbf{Ker_I}(x_i, x_j, \sigma_i) = (\sigma_i \|x_i - x_j\| + 1)^{-10} + \exp(-\sigma_i \|x_i - x_j\|)$;
- (d) Custom input and output kernels with

$$\begin{aligned}\mathbf{Ker_I}(x_i, x_j, \sigma_i) &= (\sigma_i \|x_i - x_j\| + 1)^{-10} + \exp(-\sigma_i \|x_i - x_j\|), \\ \mathbf{Ker_O}(y_i, y_j) &= (1 + \|y_i - y_j\|^2 + \|y_i - y_j\|^4)^{-1}.\end{aligned}$$

In the end, all four combinations are able to converge and produce on-par performance, in terms of distinguishing different digits. Meanwhile, each combination shows a different distribution of the output data points, which is consistent with our theoretical results.

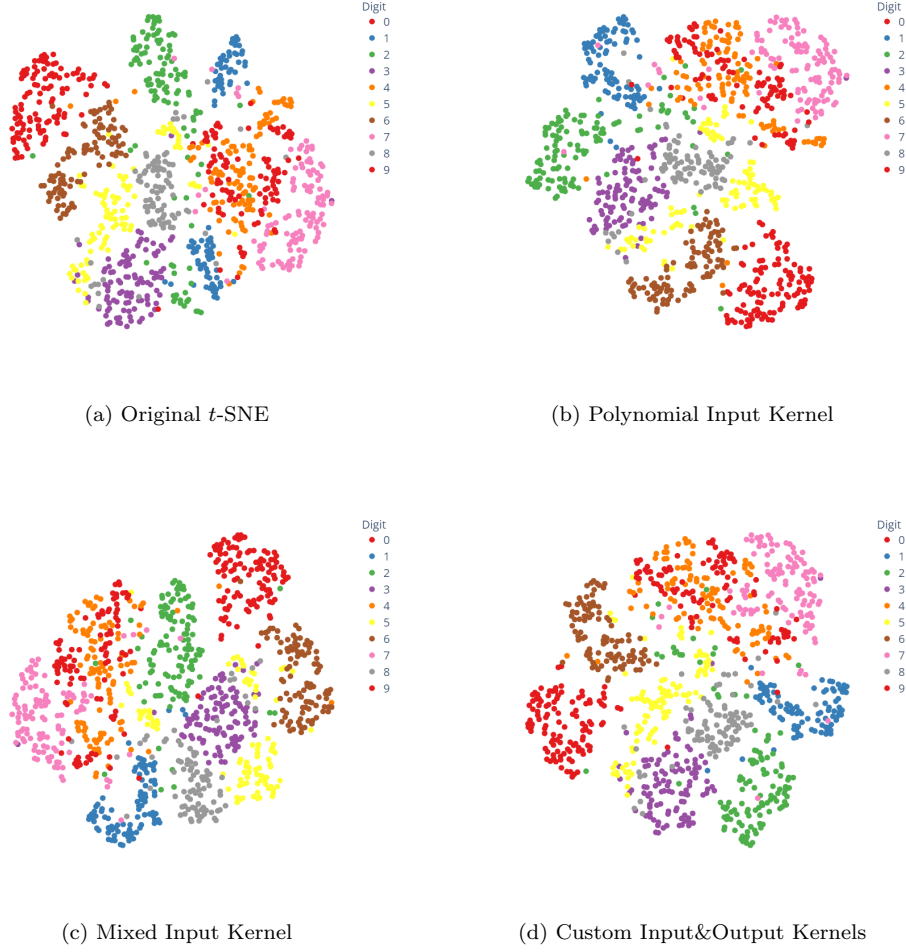


Figure 1: t -SNE ran with different combinations of input and output kernels on the MNIST data set. All three custom kernel combinations are able to perform on par with the original algorithm.

6 Existence and Uniqueness of σ^*

In this section we establish the existence and uniqueness of $\sigma_{\rho,\mu}^*(x)$. Recall that $\sigma_{\rho,\mu}^*(x)$ is the unique solution of the equation

$$F_{\rho,\mu}(x, \sigma) = \int \frac{\mathbf{Ker}_{\mathbf{I}}(x, x', \sigma)}{\int \mathbf{Ker}_{\mathbf{I}}(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_{\mathbf{I}}(x, x', \sigma)}{\int \mathbf{Ker}_{\mathbf{I}}(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho = 0 \quad (6.1)$$

Intuitively speaking, σ^* is the limit function of σ_i 's that are used to control the entropy. The purpose of this section is to prove that such limit exists and is unique for each given x . The existence and uniqueness of σ^* will become the foundation of everything we will prove afterwards. By condition 1, we can write

$$\begin{aligned} F_{\rho,\mu}(x, \sigma) &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (6.2)$$

We start with the lemma that gives F an increasing lower bound.

Lemma 1. *If μ has a continuous, bounded density $f(x)$, then there exists a constant $C > 0$ such that*

$$F_{\rho,\mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \quad (6.3)$$

Proof. Notice that for any integrable function h on $\mathbb{R}_{\geq 0}$, we have

$$\int_{\mathbb{R}^d} h(\|x\|) dx = \int_{\mathbb{S}^{d-1}} \int_0^\infty r^{d-1} h(r) dr d\nu = S_{d-1} \int_0^\infty r^{d-1} h(r) dr, \quad (6.4)$$

where $\mathbb{S}^{d-1}$ is the $(d-1)$ -dimensional sphere, $d\nu$ the corresponding surface measure, and S_{d-1} the volume of the sphere. By Condition 1, we know that

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (6.5)$$

Since w is increasing and $\lim_{t \rightarrow \infty} w(t) = \infty$, it is easy to see that

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty. \quad (6.6)$$

Denote

$$Z_d := \int_{\mathbb{R}^d} \exp(-w(\|\tau\|^\theta)) d\tau = S_{d-1} \int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt. \quad (6.7)$$

Consider

$$\begin{aligned} \int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') &= \int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx' \\ &= Z_d \sigma^{-d/\theta} \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx'. \end{aligned} \quad (6.8)$$

Because $\lim_{x \rightarrow \infty} w(x) = \infty$, the integrand of the above integral converges weakly to $\delta_x f(x')$. To prove this statement, notice that

$$\begin{aligned}
& \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx' \\
&= \frac{1}{Z_d} \int \exp\left(-w\left(\|\sigma^{1/\theta}(x - x')\|^\theta\right)\right) f(x') d\sigma^{1/\theta}(x - x') \\
&\equiv \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau.
\end{aligned} \tag{6.9}$$

Since f is bounded, $\exp(-w(\|\tau\|^\theta)) \sup f$ is integrable. By Dominant Convergence Theorem and the continuity of f , we have

$$\begin{aligned}
& \lim_{\sigma \rightarrow \infty} \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\
&= \frac{1}{Z_d} \int \lim_{\sigma \rightarrow \infty} \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\
&= f(x) \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) d\tau \\
&= f(x).
\end{aligned} \tag{6.10}$$

There exists some $\epsilon(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$ and a constant $C > 0$, that

$$\begin{aligned}
-\log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) &= -\log \left(Z_d \sigma^{-d/\theta} \right) - \log f(x) + \epsilon(\sigma) \\
&\geq C \log \sigma - \log f(x).
\end{aligned} \tag{6.11}$$

Next, since f is bounded, there exists $M > 0$ such that $f \leq M$.

$$\begin{aligned}
& \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\
&\leq M \frac{\int w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)) dx'}{\int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx'} \\
&= M \frac{\sigma^{-d/\theta} \int w(\|\sigma^{1/\theta}(x - x')\|^\theta) \exp(-w(\|\sigma^{1/\theta}(x - x')\|^\theta)) d\sigma^{1/\theta}(x - x')}{\int \exp(-w(\sigma \|(x - x')\|^\theta)) f(x') dx'} \\
&= M \frac{\sigma^{-d/\theta} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau}{\int \exp(-w(\sigma \|(x - x')\|^\theta)) f(x') dx'} \\
&\rightarrow \frac{M}{Z_d f(x)} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau < \infty.
\end{aligned} \tag{6.12}$$

The numerator converges because of condition 1. As the bound in (6.12) does not depend on σ , we can combine (6.2), (6.11) and (6.12) and deduce that

$$F_{\rho, \mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \tag{6.13}$$

□

This lemma proves the existence of $\sigma_{\rho, \mu}^*(x)$. Indeed, by (6.2), apparently

$$\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho < 0 \tag{6.14}$$

since $\rho \in (0, 1)$. Now with lemma 1, we get

$$\lim_{\sigma \rightarrow \infty} F_{\rho, \mu}(x, \sigma) = \infty. \quad (6.15)$$

By continuity of f and intermediate value theorem, $\sigma_{\rho, \mu}^*(x)$ exists. To prove the uniqueness, we are left to show $F_{\rho, \mu}$ is strictly increasing. The next lemma will do just that.

Lemma 2. *Given the same condition as in lemma 1, $F_{\rho, \mu}(x, \sigma)$ is strictly increasing in σ .*

To tackle this proof, we need one additional lemma inspired by theorem 7 in [Liu et al., 2009].

Lemma 3. *Let $f(x), g(x), h(x)$ be continuous functions $\mathbb{R}^d \rightarrow \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^d$*

$$(g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) \geq 0. \quad (6.16)$$

Let μ be an arbitrary Lebesgue measure and assume all integrals in this lemma converge. Then the inequality

$$\frac{\int_{\mathbb{R}^d} f(x) d\mu}{\int_{\mathbb{R}^d} h(x) d\mu} \geq \frac{\int_{\mathbb{R}^d} f(x) g(x) d\mu}{\int_{\mathbb{R}^d} h(x) g(x) d\mu} \quad (6.17)$$

holds. If (6.16) reverses, then (6.17) reverses.

Proof. The proof of this lemma is similar to the one in [Liu et al., 2009]. It suffices to show that

$$\int_{\mathbb{R}^d} f(x) d\mu \int_{\mathbb{R}^d} h(x) g(x) d\mu \geq \int_{\mathbb{R}^d} h(x) d\mu \int_{\mathbb{R}^d} f(x) g(x) d\mu. \quad (6.18)$$

The above inequality is equivalent to

$$\int_{\mathbb{R}^d} f(x) d\mu(x) \int_{\mathbb{R}^d} h(y) g(y) d\mu(y) \geq \int_{\mathbb{R}^d} h(x) d\mu(x) \int_{\mathbb{R}^d} f(y) g(y) d\mu(y). \quad (6.19)$$

That means we need to show

$$\mathcal{G} := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(y) (f(x) h(y) - f(y) h(x)) d\mu(x) d\mu(y) \geq 0. \quad (6.20)$$

Swap x and y in the formula of $\mathcal{G}$ and we have an equivalent expression:

$$\mathcal{G} = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(x) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y). \quad (6.21)$$

Adding the last two formulas together yields

$$\begin{aligned} 2\mathcal{G} &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (g(x) - g(y)) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(x) h(y) (g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) d\mu(x) d\mu(y) \geq 0 \end{aligned} \quad (6.22)$$

by the hypothesis of the lemma. This finishes the proof. $\square$

Proof of lemma 2. We compute the derivative of $F_{\rho, \mu}(x, \sigma)$ directly using (6.2). The following simplified notations will be used in the first several steps:

$$w = w(\sigma \|x - x'\|^\theta), \quad w' = w'(\sigma \|x - x'\|^\theta), \quad \exp(\cdot) = \exp(-w(\sigma \|x - x'\|^\theta)). \quad (6.23)$$

Omit $d\mu(x')$ in the integration for cleaner writing, and we have

$$\begin{aligned}
& \frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \\
&= - \int \frac{w' \|x - x'\|^\theta \exp(\cdot)}{\int \exp(\cdot)} - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} - \frac{(\int \exp(\cdot))'}{\int \exp(\cdot)} \\
&= - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} \quad (\text{The first and fourth terms cancel.}) \\
&= \int \frac{w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot)}{\int \exp(\cdot)} - \frac{\int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot)}{(\int \exp(\cdot))^2}. \tag{6.24}
\end{aligned}$$

We are able to move differentiation inside the integral because of dominant convergence theorem. To show that $\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \geq 0$, we just need to show

$$\int w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot) \int \exp(\cdot) \geq \int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot). \tag{6.25}$$

For each $x, y, x' \in \mathbb{R}$ if the equality

$$\begin{aligned}
& (w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta) \\
& \cdot \left(\frac{1}{w(\sigma \|y - x'\|^\theta)} - \frac{1}{w(\sigma \|x - x'\|^\theta)} \right) \geq 0 \tag{6.26}
\end{aligned}$$

holds, then by letting $f(x) = \exp(\cdot)$, $h(x) = w \cdot \exp(\cdot)$ and $g(x) = w' \cdot \|x - x'\|^\theta$, we have (6.25) to be true by lemma 3. Let $\|x - x'\|^\theta > \|y - x'\|^\theta$ without loss of generality. Then by monotonicity of w , we have $w(\sigma \|x - x'\|^\theta) > w(\sigma \|y - x'\|^\theta)$. In order for (6.26) to be true, we are left to show

$$w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta \geq 0. \tag{6.27}$$

Since $\sigma > 0$, the above inequality is equivalent to:

$$w'(\sigma \|x - x'\|^\theta) \sigma \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \sigma \|y - x'\|^\theta \geq 0. \tag{6.28}$$

Let $\tilde{w}(x) = xw'(x)$. Based on the second assumption in Condition 1 and mean value theorem, $\tilde{w}$ satisfies for all $a, b \geq 0$, $a > b$, there exists $c \in [b, a]$ that

$$\frac{aw'(a) - bw'(b)}{a - b} = w'(c) + cw''(c) \geq 0. \tag{6.29}$$

This proves (6.28) by letting $a = \sigma \|x - x'\|^\theta$ and $b = \sigma \|y - x'\|^\theta$, as desired. $\square$

We can now formally state and prove the main result of this section.

Proposition 1. *Let $\mu \in \mathcal{M}(\mathbb{R}^d)$ with C^0 density f such that f is bounded. The equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. The existence and uniqueness can be proven by combining (6.14), (6.15), lemma 1 and lemma 2. The smoothness of $\sigma_{\rho, \mu}^*(x)$ follows from implicit function theorem. $\square$

7 Uniform Convergence of $F_{\rho, \mu_n}(x, \sigma)$

As we mentioned in Section 2 and 3, we will establish the uniform convergence of $F_{\rho, \mu_n}(x, \sigma)$ under the assumption that μ_X is sub-Gaussian, instead of μ_X being compactly supported. In order to do this, we will assume Condition 3 for the input kernels. On the other hand, the reader will be able to easily verify all lemmas under with compactly supported μ_X assuming only Condition 1.

Recall that a random variable X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2\exp(-Ct^2)$ for some positive constant C . C is called the decay constant. We start by stating three lemmas from [Auffinger and Fletcher \[2023\]](#). Since they are about sub-Gaussian distributions rather than generalized input kernels, we are able to use them directly.

Lemma 4. *Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that*

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (7.1)$$

Lemma 5. *Let μ be sub-Gaussian probability measure and c_1 be the decay constant from lemma 4. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that*

$$\mathbb{P}\left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t\right) \leq Ce^{-c_1 t^2}. \quad (7.2)$$

Lemma 6. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\sigma \|x - x'\| \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ , and $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma^{-1} \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (7.3)$$

Also, let us recall that:

$$\begin{aligned} & F_{\rho, \mu}(x, \sigma) \\ &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned}$$

And $\mathcal{K}_w$ is defined as:

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0, \text{ s.t. } \lim_{n \rightarrow \infty} n^{-1/2} (\log n)^\theta \exp \left(6w \left((\tilde{C} \sqrt{\log n})^\theta \right) \right) = O(n^{-\epsilon}) \right\}. \quad (7.4)$$

Finally, to lighten up the notations, write:

$$\begin{aligned} F_{\mu_n}(x, \sigma) &= F_{\rho, \mu_n}(x, \sigma) \\ f_\sigma(x, x') &= \exp(-w(\sigma \|x - x'\|^\theta)), \\ g_\sigma(x, x') &= w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)). \end{aligned} \quad (7.5)$$

Lemma 7. *Let c_1 be the decay constant of the sub-Gaussian distribution μ , and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . If Condition 3 holds, then there exists a constant C_1 such that if $c_1 \delta^2 > C_1$, then almost surely in $X_1, \dots, X_n$,*

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \\ = O \left(n^{-1/2} \exp \left(2w \left(\left(\sqrt{\frac{8}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right) \right). \end{aligned} \quad (7.6)$$

Proof. Note that for any real numbers A, B, C, D , we have:

$$\left| \frac{A}{B} - \frac{C}{D} \right| = \left| \frac{AD - BC}{BD} \right| = \left| \frac{D(A - C) + C(D - B)}{BD} \right| \leq \left| \frac{A - C}{B} \right| + \left| \frac{C(D - B)}{BD} \right|. \quad (7.7)$$

This means:

$$\begin{aligned} \left| \frac{\int g_\sigma(x, x') d\mu_n(x')}{\int f_\sigma(x, x') d\mu_n(x')} - \frac{\int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu(x')} \right| &\leq \left| \frac{\int g_\sigma(x, x') d\mu_n(x') - \int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu_n(x')} \right| \\ &+ \left| \frac{\int g_\sigma(x, x') d\mu(x') (\int f_\sigma(x, x') d\mu_n(x') - \int f_\sigma(x, x') d\mu(x'))}{\int f_\sigma(x, x') d\mu(x') \int f_\sigma(x, x') d\mu_n(x')} \right|. \end{aligned} \quad (7.8)$$

Let $\alpha > 0$. Define

$$E'_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\left(\alpha + \frac{1}{c_1} \right) \log n} + \sqrt{d} \right\}. \quad (7.9)$$

Setting $t = \sqrt{\alpha \log n}$ in lemma 5 we have $\mathbb{P}(E'_1) \leq Cn^{-c_1 \alpha}$. To absorb the $\sqrt{d}$ term in E'_1 , we can find $\epsilon_n \rightarrow 0$ such that

$$E_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\left(\alpha + \frac{1}{c_1} + \epsilon_n \right) \log n} \right\} \quad (7.10)$$

with $\mathbb{P}(E_1) \leq Cn^{-c_1 \alpha}$. On the other hand, on E_1^c for all i, j ,

$$\|X_i - X_j\| \leq 2\sqrt{\left(\alpha + \frac{1}{c_1} + \epsilon_n \right) \log n} =: 2\gamma\sqrt{\log n}. \quad (7.11)$$

This means for $\sigma^{-1} \in [\delta^\theta, n^{\theta/2}]$ on E_1^c we have

$$\begin{aligned} \int f_\sigma(X_i, x') d\mu_n(x') &= \int \exp(-w(\sigma \|X_i - x'\|^\theta)) d\mu_n(x') \\ &\geq \exp \left(-w \left(\left(\frac{2\gamma\sqrt{\log n}}{\delta} \right)^\theta \right) \right) =: T. \end{aligned} \quad (7.12)$$

In the same way, by lemma 4, we get similar bounds

$$\int f_\sigma(X_i, x') d\mu(x') \geq (1 - Cm^{-c_1 \alpha})T \geq \frac{1}{2}T. \quad (7.13)$$

By lemma 6 with $h = f_\sigma$ and $h = g_\sigma$, there exists E_2 with $\mathbb{P}(E_2) \leq 2e^{-c\epsilon^2 n}$ such that on E_2^c ,

$$\begin{aligned} & \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| \\ & + \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \int g_\sigma(X_i, x') d\mu_n(x') - \int g_\sigma(X_i, x') d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \end{aligned} \quad (7.14)$$

Therefore on $E_1^c \cap E_2^c$, combining (7.8), (7.12), (7.13), and (7.14) gives us

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(x, x') d\mu_n(x')}{\int f_\sigma(x, x') d\mu_n(x')} - \frac{\int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu(x')} \right| \leq C\epsilon T^{-2} + O\left(\frac{\sqrt{\log n}}{n^{1/2}T^2}\right). \quad (7.15)$$

We want RHS goes to zero as $n \rightarrow \infty$. For that we need

$$\epsilon T^{-2} \rightarrow 0, \quad T^{-2} \cdot \frac{\sqrt{\log n}}{n^{1/2}} \rightarrow 0. \quad (7.16)$$

On the other hand, since $\mathbb{P}(E_1 \cap E_2) \leq Cn^{-c_1\alpha} + 2e^{-c\epsilon^2 n}$, we also need

$$c_1\alpha > 1, \quad \epsilon^2 n \rightarrow \infty. \quad (7.17)$$

Let $\epsilon = n^{-\eta}$ and by (7.16) we have $0 < \eta < 1/2$. Then, if we have $n^{-\eta}T^{-2} \rightarrow 0$ for some $\eta \in (0, 1/2)$, then we must have

$$T^{-2} \cdot \frac{\sqrt{\log n}}{n^{1/2}} \leq \frac{\sqrt{\log n}}{n^{1/2-\eta}} \rightarrow 0. \quad (7.18)$$

This means the first inequality in (7.16) implies the second. The requirements in (7.16) and (7.17) are equivalent to

$$n^{-\eta}T^{-2} \rightarrow 0, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (7.19)$$

i.e.,

$$n^{-\eta} \exp\left(w \left(\left(\frac{2\gamma\sqrt{\log n}}{\delta}\right)^\theta\right)\right) \rightarrow 0, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (7.20)$$

Let $\eta = 1/2 - \epsilon_1$ and $\alpha = 1/c_1 + \epsilon_2$. Then the above requirements are equivalent to

$$\lim_{n \rightarrow \infty} n^{-1/2+\epsilon_2} \exp\left(2w \left(\left(\sqrt{\frac{8}{(c_1+\epsilon_1)\delta^2}} \sqrt{\log n}\right)^\theta\right)\right) = 0. \quad (7.21)$$

As ϵ_1 and ϵ_2 can be arbitrarily small, the above limit can be further simplified to

$$\exists \epsilon_1, \text{ s.t. } \lim_{n \rightarrow \infty} n^{-1/2} \exp\left(2w \left(\left(\sqrt{\frac{8}{c_1\delta^2}} \sqrt{\log n}\right)^\theta\right)\right) = O(n^{-\epsilon_1}). \quad (7.22)$$

Consider the set

$$\mathcal{K}'_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \exists \epsilon_1 > 0, \text{ s.t. } \lim_{n \rightarrow \infty} n^{-1/2} \exp\left(2w \left(\left(\tilde{C}\sqrt{\log n}\right)^\theta\right)\right) = O(n^{-\epsilon_1}) \right\}. \quad (7.23)$$

Using the monotonicity of the functions involved, it is easy to show that $\mathcal{K}_w \subset \mathcal{K}'_w$, thus $\sup(\mathcal{K}'_w) \geq \sup(\mathcal{K}_w)$. By condition 3, we have $\sup \mathcal{K}'_w > 0$. Also, for any $\tilde{C} < \sup \mathcal{K}'_w$, $\tilde{C} \in \mathcal{K}'_w$. Therefore, according to (7.22), the lemma holds if

$$c_1 \delta^2 > \frac{8}{(\sup \mathcal{K}_w)^2} \geq \frac{8}{(\sup \mathcal{K}'_w)^2}. \quad (7.24)$$

We find our constant $\mathcal{C}_1$. This finishes the proof. $\square$

Using the elementary inequality $\log(x) \leq x - 1$, we can prove the next lemma in a similar way:

Lemma 8. *Let c_1 be the decay constant of the sub-Gaussian distribution μ , and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . If condition 3 holds, then there exists a constant $\mathcal{C}_2$ such that if $c_1 \delta^2 > \mathcal{C}_2$, then almost surely in $X_1, \dots, X_n$,*

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| \\ = O \left(n^{-1/2} \exp \left(w \left(\left(\sqrt{\frac{8}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right) \right) = o(1). \end{aligned} \quad (7.25)$$

Combine the above two lemmas and (6.2), we immediately obtain the following proposition:

Proposition 2. *Let μ be sub-Gaussian with decay constant c_1 from lemma 4, $\mathcal{K}_w$ be defined in (2.5) and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . $\mu_n = (1/n) \sum_i \delta_{X_i}$ is the empirical measure. Set $\delta > 0$ to be a constant such that $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$. If condition 3 holds, then there exists a constant $\epsilon > 0$, that almost surely in $X_1, \dots, X_n$,*

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = O(n^{-\epsilon}). \quad (7.26)$$

By the same method in Auffinger and Fletcher [2023] we have the following milestone proposition as well:

Proposition 3. *Suppose μ has compact support and C^1 density. If condition 3 holds, then almost surely, as $n \rightarrow \infty$,*

$$\sup_{i=1, \dots, n} |\sigma_{\mu_n}^*(X_i) - \sigma_\mu^*(X_i)| \rightarrow 0. \quad (7.27)$$

In particular,

$$|\sigma_{\mu_n}^*(X_i) - \sigma_\mu^*(X_i)| = O(|F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)|). \quad (7.28)$$

8 Proofs of the Main Theorems

8.1 Existence of μ^*

The lemmas can be borrowed directly from Auffinger and Fletcher [2023] as they irrelevant to the generalized kernels.

Lemma 9. *Any sequence in $\tilde{\mathcal{P}}_X$ is tight.*

Lemma 10. *There exists μ^* such that*

$$I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu) \quad (8.1)$$

8.2 Convergence lemmas for p_{ij}

To further lighten up the notations, define (the original definitions of f can be found in (7.5))

$$\begin{aligned}
\sigma_n(x) &= \sigma_{\rho, \mu_n}^*(x), \\
\sigma(x) &= \sigma_{\rho, \mu_X}^*(x), \\
f_n(x, x') &= f_{\sigma_n}(x, x'), \\
f(x, x') &= f_{\sigma}(x, x'), \\
p'_n(x, x') &= \frac{f_n(x, x')}{\int f_n(x, x') d\mu_n(x') - 1/n}, \quad p'(x, x') = \frac{f(x, x')}{\int f(x, x') d\mu(x')}, \\
p_n(x, x') &= \frac{1}{2}(p'_n(x, x') + p'_n(x', x)), \quad p(x, x') = \frac{1}{2}(p'(x, x') + p'(x', x)), \\
\tilde{T} &= \exp \left(w \left(\left(\sqrt{\frac{8}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right). \tag{8.2}
\end{aligned}$$

Note that $p_{j|i} = p'_n(X_i, X_j)/n$ and $p_{ij} = p_n(X_i, X_j)/n^2$, and for $\sigma^{-1} \in [\delta^\theta, n^{\theta/2}]$ we have

$$\int f(x, x') d\mu(x'), \quad \int f_n(x, x') d\mu_n(x') \geq C \tilde{T}^{-1}. \tag{8.3}$$

Lastly, we can write the definition of $\mathcal{K}_w$ as

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0, \text{ s.t. } \lim_{n \rightarrow \infty} n^{-1/2} \tilde{T}^5 = O(n^{-\epsilon}) \right\}. \tag{8.4}$$

Lemma 11. *Assuming condition 3, if $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$, then as $n \rightarrow \infty$,*

$$\sup_{i,j} |f_n(X_i, X_j) - f(X_i, X_j)| = O(n^{-\epsilon}) \tag{8.5}$$

for some $\epsilon > 0$.

Proof. By definition and assume $\sigma(x) \geq \sigma_n(x)$ WLOG (or we can always compute the other way around), we have

$$\begin{aligned}
|f_n - f| &= \left| \exp(w(\sigma(x) \|x - x\|^\theta) - w(\sigma_n(x) \|x - x\|^\theta)) - 1 \right| \exp(w(-\sigma(x) \|x - x\|^\theta)) \\
&\leq \left| \exp(w(\sigma(x) \|x - x\|^\theta) - w(\sigma_n(x) \|x - x\|^\theta)) - 1 \right| \\
&= \left| \exp(w'(\tilde{\sigma}(x) \|x - x\|^\theta)(\sigma(x) - \sigma_n(x) \|x - x\|^\theta)) - 1 \right| \\
&\leq \left| \exp(w(\tilde{\sigma}(x) \|x - x\|^\theta)(\sigma(x) - \sigma_n(x) \|x - x\|^\theta)) - 1 \right|. \tag{8.6}
\end{aligned}$$

The last step comes from (4) of condition 3. The lemma is proved if μ_X has compact support as $\exp(w(\tilde{\sigma}(x) \|x - x\|^\theta))$ is bounded from above. On the other hand, if μ_X is sub-Gaussian, let C be the constant of decay for $X_i - X_j$ and we have

$$\mathbb{P}(\|X_i - X_j\|^2 \geq \log n) \leq 2 \exp(-\log n/C^2) = 2 \exp\left(\frac{1}{c^2}\right) \cdot \frac{1}{n} \rightarrow 0. \tag{8.7}$$

This means $\|X_i - X_j\|^2 = O(\log n)$ almost surely. By (4) of condition 3 and definition of $\tilde{T}$ in (8.2), $\exp(w(\tilde{\sigma}(x) \|x - x\|^\theta)) = O(\tilde{T})$ because $\tilde{\sigma}$ sits between σ and σ_n and $\sigma_n \rightarrow \sigma$.

We turn to the rest of terms in (8.6). If μ_X has compact support, then the lemma is proved by simply applying proposition 3. Finally, we know that

$$\sigma(x) - \sigma_n(x) = O(F_{\rho, \mu}(x, \sigma) - F_{\rho, \mu}(x, \sigma_n)) = O(n^{-1/2} \tilde{T}^2). \tag{8.8}$$

Applying (8.8) and by the simple limit $\lim_{x \rightarrow 0} (e^x - 1)/x = 1$, the difference (8.6) can be bounded as

$$|f_n - f| = O(n^{-1/2}(\log n)^\theta \tilde{T}^3). \quad (8.9)$$

The rest of the proof can be finished by constructing a similar set as in (7.23) and using the same argument as in the proof of lemma 7. (Simply compare the formula in (8.9) with (8.4)). $\square$

Lemma 12. *Assuming condition 3 and μ_X is sub-Gaussian, if $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$, then there exists $\epsilon > 0$ such that as $n \rightarrow \infty$,*

$$\begin{aligned} \sup_{i,j} |p'_n(X_i, X_j) - p'(X_i, X_j)| &= O(n^{-\epsilon}), \\ \sup_{i,j} |p_n(X_i, X_j) - p(X_i, X_j)| &= O(n^{-\epsilon}). \end{aligned}$$

Proof. Once again, we omit X_i and X_j for lighter notations. Using the triangular inequality, we have

$$\begin{aligned} |p'_n - p'| &= \left| \frac{f_n}{\int f_n d\mu_n - 1/n} - \frac{f}{\int f d\mu} \right| \\ &\leq \left| \frac{f_n}{\int f_n d\mu_n - 1/n} - \frac{f_n}{\int f_n d\mu_n} \right| + \left| \frac{f_n}{\int f_n d\mu_n} - \frac{f_n}{\int f d\mu_n} \right| \\ &\quad + \left| \frac{f_n}{\int f d\mu_n} - \frac{f_n}{\int f d\mu} \right| + \left| \frac{f_n}{\int f d\mu} - \frac{f}{\int f d\mu} \right|. \end{aligned} \quad (8.10)$$

In order to get the convergence, we need to check the convergence of each term. Note that $f, f_n \leq 1$. Denote The first can be bounded by using (7.12) and (7.22):

$$\left| \frac{f_n}{\int f_n d\mu_n - 1/n} - \frac{f_n}{\int f_n d\mu_n} \right| \leq \frac{1}{n} \left| \frac{1}{\int f_n d\mu_n - 1/n} \cdot \frac{1}{\int f_n d\mu_n} \right| = O(n^{-1} \tilde{T}^2). \quad (8.11)$$

Recall the definitions of $\mathcal{K}_w$ and $\tilde{T}$, we have $O(n^{-1} \tilde{T}^2) = O(n^{-\epsilon})$ if $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$.

Continue to the second term, we compute

$$\left| \frac{f_n}{\int f_n d\mu_n} - \frac{f_n}{\int f d\mu_n} \right| \leq \frac{|f_n - f|}{|\int f_n d\mu_n| |\int f d\mu_n|} = O(|f_n - f| \tilde{T}^2) = O(n^{-1/2}(\log n)^\theta \tilde{T}^5) \quad (8.12)$$

by lemma 11. This term is also $O(n^{-\epsilon})$ if $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$, as the RHS of (8.12) decays faster than the formula in (8.4).

For the third them we can use lemma 6 and repeat the procedure that leads us to (7.15), which gives us

$$\left| \frac{f_n}{\int f d\mu_n} - \frac{f_n}{\int f d\mu} \right| \leq \frac{|\int f d\mu_n - \int f d\mu|}{|\int f d\mu| |\int f d\mu_n|} = O(n^{-1/2} \tilde{T}^2), \quad (8.13)$$

which is also $O(n^{-\epsilon})$ based on the hypothesis of the lemma.

Finally, we have a similar inequality as the second term for the fourth term:

$$\left| \frac{f_n}{\int f d\mu} - \frac{f}{\int f d\mu} \right| = O(|f_n - f| \tilde{T}). \quad (8.14)$$

The RHS is $O(n^{-\epsilon})$ because it decays faster than the RHS of (8.12). The proof is finished by putting (8.10), (8.11), (8.12), (8.13), and (8.14) together. $\square$

Lemma 13. *Assuming condition 3 and μ_X is sub-Gaussian, if $c_1\delta^2 > 8/(\sup \mathcal{K}_w)^2$, then there exists $\epsilon > 0$ such that as $n \rightarrow \infty$,*

$$\sup_{i,j} |p_n(X_i, X_j) \log p_n(X_i, X_j) - p(X_i, X_j) \log p(X_i, X_j)| = O(n^{-\epsilon}).$$

Proof. Let $\epsilon_n = n^{-1/2}(\log n)^\theta \tilde{T}^5$, where we inherit from (8.12). We compute

$$\begin{aligned} p_n \log p_n &= p \log p + p \log \left(\frac{p_n}{p} \right) + (p_n - p) \log p_n \\ &= p \log p + p \log \left(\frac{p + O(\epsilon_n)}{p} \right) + O(\epsilon_n) \log p_n. \end{aligned} \quad (8.15)$$

The third term will apparently vanish. For the second term, recall that $p \leq \tilde{T}^{-1}$. We want

$$O(\epsilon_n)/p = O(\epsilon_n \tilde{T}) = O(n^{-1/2}(\log n)^\theta \tilde{T}^6) = O(n^{-\epsilon}), \quad (8.16)$$

which is just enough by (8.4) and condition 3. $\square$

Lemma 14. *Assuming condition 3 and μ_X is sub-Gaussian, if $c_1\delta^2 > 8/(\sup \mathcal{K}_w)^2$, then as $n \rightarrow \infty$,*

$$\sum_j p_{j|i} \log p_{j|i} + \log(\rho n) = o(1). \quad (8.17)$$

Proof. Recall that $p_{j|i} = \frac{1}{n} p'_n(X_i, X_j)$. Therefore

$$\begin{aligned} \sum_j p_{j|i} \log p_{j|i} &= \sum_{j \neq i} \frac{1}{n} p'_n(X_i, X_j) \log \left(\frac{1}{n} p'_n(X_i, X_j) \right) + \log n + \log \rho \\ &= \frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \log p'_n(X_i, X_j) + \log \rho + \log n \left(1 - \frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \right) \\ &= \frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \log p'_n(X_i, X_j) + \log \rho \end{aligned} \quad (8.18)$$

because by definition $n - \sum_{j \neq i} p'_n(X_i, X_j) = 0$. We proceed to write the first summation as an integral over the measure μ_n , with the difference being the diagonal terms.

$$\begin{aligned} &\frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \log p'_n(X_i, X_j) - \int p'_n(X_i, x') \log p'_n(X_i, x') d\mu_n(x') \\ &= -\frac{1}{n} \frac{1}{\int f(X_i, x') d\mu_n(x') - \frac{1}{n}} \log \left(\frac{1}{\int f(X_i, x') d\mu_n(x') - \frac{1}{n}} \right). \end{aligned} \quad (8.19)$$

If μ_X has compact support, then the above diagonal term is $O(n^{-1})$. If μ_X is sub-Gaussian, then by (8.3), the diagonal term in (8.19) is $O(n^{-1} \tilde{T} \log(\tilde{T})) = o(1)$ by condition 3 and the definition of $\mathcal{K}_w$. The rest of convergence follows in the exact same way in Auffinger and Fletcher [2023]. $\square$

8.3 Lower Bound for $\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y)$

We write $g(y, y') = \mathbf{Ker}_O(y, y')$ from now on for simpler notations.

Lemma 15. *There exists a set of points $y^*(X)$ such that*

$$\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y) = L_{n,\rho}(X, y^*). \quad (8.20)$$

Proof. Let $k \in \{1, \dots, n\}$. We are going to show that

$$\lim_{|y_k| \rightarrow \infty} L_{n,\rho}(X, y) = \infty. \quad (8.21)$$

By definition, L is the relative entropy and

$$L_{n,\rho}(X, y) = \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (8.22)$$

For n given and fixed, it suffices to show that

$$\sum_{i \neq j} p_{ij} \log(q_{ij}) = \sum_{i \neq j} p_{ij} \log \left(\frac{g(y_i, y_j)}{\sum_{m \neq l} g(y_m, y_l)} \right) \rightarrow -\infty. \quad (8.23)$$

Recall that $g(y, y')$ is bounded. We have

$$\begin{aligned} & - \sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq l} g(y_m, y_l) \right) = - \log \left(\sum_{m \neq l} g(y_m, y_l) \right) \\ & \leq - \sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq k, l \neq k, m \neq l} g(y_m, y_l) \right) < \infty. \end{aligned} \quad (8.24)$$

On the other hand, we have $g(y_i, y_k) = k(\|y_i - y_k\|) \rightarrow 0$ as $|y_k| \rightarrow \infty$ because $\|y_i - y_k\| \rightarrow \infty$ and k satisfies

$$\iint_{(0, \infty)^2} k(\|x - x'\|) dx dx' < \infty. \quad (8.25)$$

This means

$$\lim_{|y_k| \rightarrow \infty} \sum_{i \neq j} p_{ij} \log(g(y_i, y_j)) \rightarrow -\infty. \quad (8.26)$$

Hence, the minimum of the $L_{n,\rho}(X, y)$ is achieved on a bounded set, the lemma follows by the continuity of $L_{n,\rho}(X, y)$. $\square$

Lemma 16. *Assuming condition 3, as $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} p_{ij} \log p_{ij} = \iint p(x, x') \log p(x, x') d\mu d\mu - 2 \log n + o(1). \quad (8.27)$$

Proof. First recall that $p_{ij} = p_n(X_i, X_j)/n^2$. Therefore we have

$$\begin{aligned} & \sum_{i \neq j} p_{ij} \log p_{ij} = \sum_{i \neq j} \frac{1}{n^2} p_n(X_i, X_j) \log \left(\frac{1}{n^2} p_n(X_i, X_j) \right) \\ & = \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - 2 \log n \sum_{i \neq j} \frac{1}{n^2} p_n(X_i, X_j) \\ & = \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - 2 \log n, \end{aligned} \quad (8.28)$$

because the second summation is just the sum of all probability mass. Now we get the $2 \log n$ term and we are left to deal with the first summation. We begin by showing that the diagonal terms do not matter in our estimation. consider

$$\begin{aligned} & \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - \iint p_n(x, x') \log p_n(x, x') d\mu_n(x) d\mu_n(x') \\ &= -\frac{1}{n^2} \sum_{t=1}^n \frac{1}{\int f(X_t, x') d\mu_n(x') - \frac{1}{n}} \log \left(\frac{1}{\int f(X_t, x') d\mu_n(x') - \frac{1}{n}} \right). \end{aligned} \quad (8.29)$$

If we refer to the proof of lemma 14 in [Auffinger and Fletcher \[2023\]](#), we can see that the difference of diagonal term vanishes and it does not depend on the specific X_i . Therefore in our proof, when we sum over all the X_i and normalize by dividing n^2 , we reach the exact same conclusion. The diagonal term is $O(n^{-2} \tilde{T} \log \tilde{T}) = O(n^{-\epsilon})$ which is $o(1)$ if $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$.

Then we apply lemma 13 and since $d\mu_n \times d\mu_n$ is a probability measure, we have

$$\iint p_n(x, x') \log p_n(x, x') d\mu_n(x) d\mu_n(x') = \iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') + o(1), \quad (8.30)$$

which immediately implies

$$\sum_{i \neq j} p_{ij} \log p_{ij} = \iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') - 2 \log n + o(1). \quad (8.31)$$

The proof will be finished if we are able to replace the measure in the last integral with $d\mu$. We verify that

$$h(x, x', \sigma) := p(x, x') \log p(x, x') \quad (8.32)$$

satisfies all the requirements in lemma 6. First μ is sub-Gaussian by construction. Then is easy to verify that h is C^1 for all x, x' and σ . Moreover, by calculus we have $p(x, x') \in (0, 1]$ and $\lim_{a \rightarrow 0} a \log a = 0$. Thus $p \rightarrow 0$ as $\|x - x'\| \sigma \rightarrow \infty$. It is also obvious that h is bounded and $h \rightarrow 0$ as $\|x - x'\| \sigma \rightarrow \infty$. The rest of the conditions are also very easy to prove. We can finally conclude that

$$\iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') = \iint p(x, x') \log p(x, x') d\mu(x) d\mu(x') + o(1). \quad (8.33)$$

Combine this estimation with (8.31) and we can conclude

$$\sum_{i \neq j} p_{ij} \log p_{ij} = \iint p(x, x') \log p(x, x') d\mu(x) d\mu(x') - 2 \log n + o(1), \quad (8.34)$$

as desired. □

Before we move on, we would like to take a small deviation and compute the partial derivative of the relative entropy. This is important because it will impact the property of the set $\tilde{\mathcal{P}}_X$.

Lemma 17. *Let $X \in (\mathbb{R}^d)^n$ and $Y = (y_1, \dots, y_n) \in (\mathbb{R}^s)^n$ be the input and output of the t -SNE algorithm, then we have*

$$\frac{\partial L_{n,p}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \quad (8.35)$$

Proof. By definition,

$$\begin{aligned} L_{n,\rho}(X, Y) &= \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \\ &= \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \end{aligned} \quad (8.36)$$

The first summation does not depend on y_i and will vanish when taking the partial derivative. Moreover, we can further breakdown the second summation:

$$\begin{aligned} - \sum_{i \neq j} p_{ij} \log q_{ij} &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \sum_{i \neq j} p_{ij} \log \left(\sum_{k \neq l} g(y_k, y_l) \right) \\ &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \log \left(\sum_{k \neq l} g(y_k, y_l) \right). \end{aligned} \quad (8.37)$$

Taking derivative of the last expression and note that only terms that contain y_i remains. This means

$$\begin{aligned} \frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{1}{\sum_{k \neq l} g(y_k, y_l)} \frac{\partial}{\partial y_i} g(y_i, y_j) \right) \\ &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{g(y_i, y_j)}{\sum_{k \neq l} g(y_k, y_l)} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} \right) \\ &= - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \end{aligned} \quad (8.38)$$

□

We can easily do a sanity test by letting $g(y_i, y_j) = (1 + \|y_i - y_j\|^2)^{-1}$ and get formula (A.1) in [Auffinger and Fletcher \[2023\]](#). The derivative in lemma 17 explains why $\tilde{\mathcal{P}}_X$ has that specific form. We are gradually marching towards the end. Onto the next lemma we go:

Lemma 18. *Assume condition 3 and μ_X is sub-Gaussian. The following holds:*

$$\liminf_{n \rightarrow \infty} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \geq \iint p(x, x') \log \frac{p(x, x')}{q(y, y')} d\mu d\mu. \quad (8.39)$$

Proof. Just as in lemma 17, we first break down the summation:

$$\sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (8.40)$$

The convergence of the first summation has been proved in lemma 16. We just need to deal with the second summation. Note that we now have

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i} \quad (8.41)$$

if y_i is involved. By the definition of q_{ij} , we have

$$\begin{aligned}
& - \sum_{i \neq j} p_{ij} \log(q_{ij}) \\
&= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log g(y_i, y_j) + \log \left(\frac{1}{n^2} \sum_{i \neq j} g(y_i, y_j) \right) + 2 \log n \\
&= \iint p_n(x, x') g(y, y') d\mu_n d\mu_n + \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) + 2 \log n. \tag{8.42}
\end{aligned}$$

By Fatou's lemma and lemma 12, we have

$$\liminf_{n \rightarrow \infty} \iint p_n(x, x') \log g(y, y') d\mu_n d\mu_n \geq \iint p(x, x') \log g(y, y') d\mu d\mu. \tag{8.43}$$

Similarly, using Fatou lemma again and we have

$$\liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu d\mu. \tag{8.44}$$

The $1/n$ is the duplicated diagonal term in the empirical measure and will obviously vanish as we take $n \rightarrow \infty$. Moreover, because of our assumption on the kernel g , the integrals on the RHS of the last two inequalities converge. In addition, since $\log$ is a continuous and increasing function, (8.44) implies

$$\liminf_{n \rightarrow \infty} \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) \geq \log \left(\iint g(y, y') d\mu d\mu \right). \tag{8.45}$$

The proof is finished by combining (8.42), (8.43), (8.45) and lemma 16. The $2 \log n$ cancelled out because of (8.31). $\square$

Proposition 4. *There exists a sub-sequence $n(k)$ such that*

$$\liminf_{k \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^{n(k)}} L_{n(k), \rho}(X, Y) \geq I_\rho(\mu). \tag{8.46}$$

Proof. Let y^* be given in the lemma and define

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i}. \tag{8.47}$$

We can easily show that μ_n converges weakly to a measure μ . By lemma 24 in section 10, we know that the limiting measure $\mu \in \tilde{\mathcal{P}}_X$. The proof is complete by using lemma 15 and lemma 18. $\square$

8.4 Upper Bound for $\inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y)$

For convenience, define

$$q_n(y, y') = \frac{g(y, y')}{\iint g(y, y') d\mu_n(y) d\mu_n(y') - \frac{1}{n} k(0)}. \tag{8.48}$$

Here $k(0) = g(y, y)$ is the value of the output kernel when both inputs are equal. Since multiplying a constant to g does not change anything, we can just treat $k(0) = 1$ without loss of generality as everything is normalized. By lemma 10 there exists μ^* such that $I(\mu^*) = \inf_{\mu} I_\rho(\mu)$. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. random variables drawn from $\mu \mu^*$ and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ their empirical measure. μ_n converges to μ^* weakly.

Lemma 19. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \iint p_n(x, x') \log \left(\frac{p_n(x, x')}{q_n(y, y')} \right) d\mu_n d\mu_n + o(1). \quad (8.49)$$

Proof. We begin with the familiar computation

$$\begin{aligned} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) \\ &= \iint p_n(x, x') \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) d\mu_n d\mu_n + \text{Err}_n, \end{aligned} \quad (8.50)$$

where

$$\begin{aligned} \text{Err}_n &= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \left(\log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) - \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right) \right) \\ &= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) \\ &\quad + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \end{aligned} \quad (8.51)$$

By (8.7), we know that $\|X_i - X_j\| \leq \log n$ almost surely as $n \rightarrow \infty$. By bounds in (8.3) and condition 3, it is easy to see that the first summation in the error term is $o(1)$ and we can write

$$\text{Err}_n = o(1) + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \quad (8.52)$$

For the second term, it is already clear that

$$\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} = o(1). \quad (8.53)$$

On the other hand, we have by Fatou's lemma:

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k \neq l} g(Y_k, Y_l) = \liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu^* d\mu^* < \infty, \quad (8.54)$$

by our assumption on the output kernel g . We can see that the second term in Err_n is also $o(1)$, which immediately implies $\text{Err}_n = o(1)$ and completes the proof. $\square$

The next lemma is needed to closing up the lower bound part of the main theorem.

Lemma 20. *As $n \rightarrow \infty$, we have*

$$\iint g(y, y') d\mu_n(y) d\mu_n(y) - \frac{1}{n} = \iint g(y, y') d\mu^* d\mu^* + o(1). \quad (8.55)$$

Proof. Since $\mu_n \rightarrow \mu$ weakly, it is tight by Prokhorov's theorem. For arbitrary $\epsilon > 0$, there exists a compact set K such that for all n , $\mu^*(K^c), \mu_n(K^c) \leq \epsilon$. Since $g(y, y')$ is bounded, there exists a constant C such that

$$\begin{aligned} & \left| \iint g(y, y') d\mu_n(y) d\mu_n(y) - \iint g(y, y') d\mu^* d\mu^* \right| \\ &= \left| \iint g(y, y') \mathbb{1}_{K \times K} (d\mu_n d\mu_n - d\mu^* d\mu^*) \right| + C\epsilon. \end{aligned} \quad (8.56)$$

Define

$$G_n(y) = \int g(y, y') \mathbb{1}_K d\mu_n(y'), \quad G(y) = \int g(y, y') \mathbb{1}_K d\mu^*(y'). \quad (8.57)$$

We now require the kernel g to have bounded derivative, and we have the following claims:

Claim 1. $G_n(y)$ is uniformly equicontinuous on K .

Proof of Claim. By mean value theorem and condition 2, we get

$$\begin{aligned} |G_n(y) - G_n(\tilde{y})| &= \left| \int k(\|y - y'\|) - k(\|\tilde{y} - y'\|) \mathbb{1}_K d\mu_n(y') \right| \\ &\leq \int \|y - \tilde{y}\| \left| \frac{k(\|y - y'\|) - k(\|\tilde{y} - y'\|)}{\|(y - y') - (\tilde{y} - y')\|} \right| \mathbb{1}_K d\mu_n(y') \\ &= O(\|y - \tilde{y}\|). \end{aligned} \quad (8.58)$$

□

Claim 2. $\sup_{y \in K} |G_n(y) - G(y)| = o(1)$.

Proof of Claim. Since K is compact, it suffices to show that for each $y \in K$, there exists $\delta > 0$ such that

$$\sup_{\tilde{y} \in B(y, \delta)} |G_n(\tilde{y}) - G(\tilde{y})| \leq \epsilon. \quad (8.59)$$

Towards this end, by triangular inequality,

$$|G_n(\tilde{y}) - G(\tilde{y})| \leq |G_n(\tilde{y}) - G_n(y)| + |G_n(y) - G(y)| + |G(y) - G(\tilde{y})|. \quad (8.60)$$

By the previous claim, the first term on the RHS vanishes because of equicontinuity. The second term vanishes because of weak convergence and the boundedness of the integrand $g(y, y')$. Finally, the last term vanishes because we can prove the equicontinuity of G using a similar argument like the claim 1.

□

By claim 2 we have

$$\iint g(y, y') \mathbb{1}_{K \times K} d\mu_n d\mu_n = \iint g(y, y') \mathbb{1}_{K \times K} d\mu^* d\mu_n + o(1) = \int G(y) \mathbb{1}_K d\mu_n(y) + o(1). \quad (8.61)$$

Since G is bounded and continuous by condition 2, using weak convergence of μ_n and we have

$$\int G(y) \mathbb{1}_K d\mu_n(y) = \int G(y) \mathbb{1}_K d\mu^*(y) + o(1) = \iint g(y, y') d\mu^* d\mu^* + o(1), \quad (8.62)$$

as desired.

□

Lemma 21. *If μ^* has compact support in Y , then*

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p_n(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (8.63)$$

Proof. By compactness, $\|y - y'\|$ is bounded from above and $g(y, y')$ does not decay to zero. In that case, by lemma 12, we can replace the p_n in the integral by p :

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p(x, x') \log(g(y, y')) d\mu_n d\mu_n + o(1). \quad (8.64)$$

mimicking the proof of lemma 16 we can show that

$$\iint p(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (8.65)$$

This finishes the proof. □

Lemma 22. *If μ^* has compact support in Y , then*

$$\sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \iint p(x, x') \log \left(\frac{p(x, x')}{q(y, y')} \right) d\mu^* d\mu^* + o(1) \quad (8.66)$$

Proof. This lemma can be proved by combining the last 3 lemmas and lemma 16. □

Using the previous lemma and we can immediately get

Proposition 5. *If μ^* has compact support in Y , then*

$$\limsup_{n \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y) \leq I_\rho(\mu^*). \quad (8.67)$$

8.5 Proof of theorem 1

Write

$$d_{n, \rho}(X) = \inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y). \quad (8.68)$$

By proposition 5, 4 and the fact that $\mu^* = \arg \min_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$, there exists $\mu \in \tilde{\mathcal{P}}_X$ and a subsequence $n(k)$ such that

$$I_\rho(\mu^*) \geq \limsup_{n \rightarrow \infty} d_{n, \rho}(X) \geq \limsup_{n \rightarrow \infty} d_{n(k), \rho}(X) \geq I_\rho(\mu) \geq I_\rho(\mu^*). \quad (8.69)$$

Therefore the limit

$$\lim_{n \rightarrow \infty} d_{n(k), \rho}(X) = I_\rho(\mu) = I_\rho(\mu^*) \quad (8.70)$$

holds when we look at the sub-sequence $n(k)$ with the empirical measure μ converging to μ weakly. For the general limit, we can repeat the reasoning above by looking at any convergent sub-sequences of $d_{n, \rho}(X)$. We are able to find a sub-sequence inside each convergent sub-sequence and they all converges to $I_\rho(\mu^*)$. Therefore each convergent sub-sequence of $d_{n, \rho}(X)$ has the same limit and the theorem is proved. □

8.6 Proof of theorem 2

We need another lemma before proving this theorem.

Lemma 23. *For every $\mu \in \tilde{\mathcal{P}}_X$, there exists a function $h_\mu(x)$ that does depend on y such that*

$$\int p(x, x') \log(1 + \|y - y'\|) d\mu = h_{\mu(x)} - \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu}. \quad (8.71)$$

Moreover, if μ_X has compact support, then h_μ is continuous on μ_X .

Proof. Recall from (17), we have

$$\begin{aligned} 0 &= \int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') \\ &= \int p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu - \frac{\int \frac{\partial}{\partial y} g(y, y') d\mu}{\iint g(y, y') d\mu d\mu} \\ &= \nabla_y \left(\int p(x, x') \log(g(y, y')^{-1}) d\mu - \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu} \right). \end{aligned} \quad (8.72)$$

This means that there exists $h_\mu : \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$h_{\mu(x)} = \int p(x, x') \log(g(y, y')^{-1}) d\mu + \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu}. \quad (8.73)$$

Since $h_{\mu(x)}$ does not change with y , it suffices to show that $\int p(x, x') \log(g(y, y')^{-1}) d\mu$ is continuous in x . Let $\epsilon > 0$ and suppose $x_n \rightarrow x$. Since p is continuous, and μ_X has compact support, we have for some constant C ,

$$\iint \log(g(y, y')^{-1}) d\mu d\mu \leq C \int p(x, x') \log(g(y, y')^{-1}) d\mu < \infty. \quad (8.74)$$

In particular, for almost every y , $\int \log(g(y, y')^{-1}) d\mu < \infty$. Since $p(x, x')$ is continuous and μ_X has compact support, $p(x, x')$ is uniformly continuous. Specifically, there exists $\delta = \delta(y, \mu)$ such that if $\|x_n - x\| \leq \delta$,

$$\sup_x |p_n(x, x') - p(x, x')| \leq \frac{\epsilon}{\log(g(y, y')^{-1}) d\mu}. \quad (8.75)$$

This means

$$\left| \int (p_n(x, x') - p(x, x')) \log(g(y, y')^{-1}) d\mu \right| < \epsilon, \quad (8.76)$$

which is the definition of continuity. The proof is finished. $\square$

Proof of Theorem 2. We prove by contradiction. Define $D_n = \{\|y\| \geq n\}$. If μ^* does not have compact support in Y then $\mu(D_n) > 0$ for all n . By lemma 23, there exists continuous $h(x)$ such that

$$\int p(x, x') \log(g(y, y')^{-1}) d\mu^* = h(x) - \frac{\int g(y, y') d\mu^*}{\iint g(y, y') d\mu^* d\mu^*}. \quad (8.77)$$

Multiply both sides by $\mathbb{1}_{D_n(x,y)}/\mu^*(D_n)$ and integrate with respect to the other variable:

$$\begin{aligned} & \iint p(x, x') \log(g(y, y')^{-1}) \frac{\mathbb{1}_{D_n(x,y)}}{\mu^*(D_n)} d\mu^* d\mu^* \\ &= \int h(x) \frac{\mathbb{1}_{D_n(x,y)}}{\mu^*(D_n)} d\mu^* - \frac{\iint g(y, y') \frac{\mathbb{1}_{D_n(x,y)}}{\mu^*(D_n)} d\mu^* d\mu^*}{\iint g(y, y') d\mu^* d\mu^*}. \end{aligned} \quad (8.78)$$

Since h is continuous and μ_X has compact support, the first term of the RHS is bounded uniformly in n . Moreover, $g(y, y')$ is bounded so the second term is also bounded uniformly in n . It follows that the LHS is bounded in uniformly in n . With this being said, because $p(x, x') > 0$ and μ_X has compact support, there exists a constant c such that $p(x, x') \geq c$ for all x, x' in the support of μ_X .

Meanwhile, since the output kernel $g(y, y') = k(\|x - x'\|)$ is integrable, it has a decaying tail and there exists $C(n)$ such that $C(n) \rightarrow \infty$ and if $\|x - x'\| \geq n/2$, $g(y, y')^{-1} = k(\|x - x'\|)^{-1} \geq \log(1 + C(n))$. We compute

$$\begin{aligned} & \frac{1}{\mu^*(D_n)} \iint p(x, x') \log(g(y, y')^{-1}) \mathbb{1}_{\|y\| \geq n} d\mu^* d\mu^* \\ & \geq \frac{1}{\mu^*(D_n)} \iint p(x, x') \log(g(y, y')^{-1}) \mathbb{1}_{\|y\| \geq n} \mathbb{1}_{\|y - y'\| \geq n/2} d\mu^* d\mu^* \\ & \geq c \frac{\log(1 + C(n))}{\mu^*(D_n)} \iint \mathbb{1}_{\|y\| \geq n} \mathbb{1}_{\|y'\| \leq n/2} d\mu^* d\mu^* \\ & = c \log(1 + C(n)) \mu^*(\|Y'\| \leq n/2). \end{aligned} \quad (8.79)$$

Clearly,

$$\lim_{n \rightarrow \infty} c \log(1 + C(n)) \mu^*(\|Y'\| \leq n/2) = \infty. \quad (8.80)$$

This contradicts with the boundedness. We must have that $\mu^*(D_n) = 0$ for n sufficiently large. That is, $\mu^*(\|Y\| \geq n) = 0$, so μ^* has compact support in Y . $\square$

9 Conclusion

We studied the asymptotic behavior and the equilibrium measures of the t-SNE algorithm with generalized input and output kernels. Our results significantly advances the understanding of t-SNE by investigating its asymptotic behavior and equilibrium measures under generalized input and output kernels. Our key contributions can be summarized as follows:

1. **A Foundation for Generalized Kernels:** We established a concrete, yet general, formulation for both input and output kernels, providing a solid foundation for subsequent theoretical analyses.
2. **Guaranteed Convergence:** We identified specific conditions on these kernels that ensure the existence of an equilibrium measure, μ^* , and guarantee the convergence of the perplexity-controlled t-SNE algorithm. These conditions provide crucial insights into the essential properties of kernels required for optimal algorithmic performance.
3. **Relaxed Constraints and Novel Proofs:** Through the development of novel proof techniques, we demonstrated the existence and uniqueness of the perplexity-controlling parameter, σ^* , while simultaneously relaxing the constraints on the underlying probability measure, μ_X . This significantly broadens the applicability of t-SNE and enables a more comprehensive convergence analysis.

4. **Extended Convergence Analysis:** We rigorously proved the existence of the equilibrium measure, μ^* , and demonstrated that, under the specified conditions, the output of the generalized t-SNE algorithm converges to this equilibrium. This result dramatically extends the convergence analysis in Auffinger and Fletcher [2023] to a significantly wider class of kernels.

5. **Empirical Validation through Simulation:** We performed extensive simulations across a diverse range of kernel combinations, providing strong empirical support for the validity and robustness of our theoretical findings.

These advancements pave the way for more flexible and powerful applications of t-SNE in various domains. Future work may focus on designing novel kernels based on this framework, further optimizing the algorithm's performance and extending its capabilities.

10 Auxiliary lemmas

Recall that $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i, Y_i}$ is the empirical measure of both the input and output. We just ignore the first or the second part based on the integration variable because it is a probability measure. Let $Y = Y^* = (Y_1, \dots, Y_n)$ be a minimizer of L . Recall from (8.38) that

$$\frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} = 0. \quad (10.1)$$

Lemma 24. *Suppose that μ_X has either compact support and a C^1 density $f(x)$ or μ_X has sub-Gaussian tails. Then for μ -a.e. x and y we have*

$$\int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (10.2)$$

Proof. Let $\alpha(x, y)$ be a test function. It suffices to show that

$$\iint \alpha(x, y) (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (10.3)$$

Recalling the definition of p_n and q_n , we see from (8.38) that for all i ,

$$\frac{1}{n^2} \sum_{j=1, j \neq i}^n (p_n(X_i, X_j) - q_n(X_i, X_j)) \frac{\frac{\partial}{\partial Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)} = 0. \quad (10.4)$$

Multiply by $\alpha(X_i, Y_i)$ for each i and the both sides are still zero. Afterwards, take summation in i and we get

$$\begin{aligned} & \iint \alpha(x, y) (p_n(x, x') - q_n(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \frac{1}{n^2} \sum_{1 \leq i, j \leq n} (p_n(X_i, X_j) - q_n(X_i, X_j)) \frac{\frac{\partial}{\partial Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)} = 0 \end{aligned} \quad (10.5)$$

if we require $k'(0) = 0$. By lemma 12 and lemma 20, we have

$$p_n(X_i, X_j) - q_n(Y_i, Y_j) = p(X_i, X_j) - q(Y_i, Y_j) + o(1). \quad (10.6)$$

Since both α and $\frac{\partial}{\partial y_i} g(y_i, y_j)$ are bounded based on our assumptions, we have

$$\begin{aligned} & \iint \alpha(x, y)(p_n(x, x') - q_n(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n + o(1). \end{aligned} \quad (10.7)$$

By lemma 25 and lemma 26 below we have

$$\begin{aligned} & \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu d\mu + o(1). \end{aligned} \quad (10.8)$$

Since α is an arbitrary function,

$$\int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (10.9)$$

□

Lemma 25. *The following holds as $n \rightarrow \infty$:*

$$\iint \alpha(x, y)p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n = \iint \alpha(x, y)p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu d\mu + o(1) \quad (10.10)$$

This lemma can be proved by mimicking lemma 16

Lemma 26. *The following holds as $n \rightarrow \infty$:*

$$\iint \alpha(x, y)q(y, y') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n = \iint \alpha(x, y)q(y, y') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu d\mu + o(1) \quad (10.11)$$

This lemma can be proved by mimicking lemma 19.

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